06 — EU AI Act & Security Review (Mission 04)

Station: AI Security Officer
Mission: 04 | **Squadron:** Colby | **Officer:** Mubina Rahaman Jerin
Scope: `01–04` (DPO loss core, implicit rewards, SimPO, activation steering & alignment audit)
Legal basis: Regulation (EU) 2024/1689 (EU AI Act), as amended by the Digital Omnibus
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1. Classification

Question	Answer
Is this code an "AI system" (Art. 3(1))?	**Not in isolation.** The modules in Mission 04 are algorithmic optimization components (loss functions, preference evaluators, hooks, and audit benchmarks) tested on a small research model (AURA Lab, 350k parameters, synthetic language). They are not placed on the market and serve no end users.
Would AURA utilizing Subsystem 04 be in scope?	**Yes.** A deployed conversational assistant aboard the *USSE Noosphere* utilizing these alignment modules constitutes an AI system under Art. 3(1).
What risk tier applies to AURA?	Under Annex III, general conversational assistants are not classified as high-risk unless they control safety components of critical physical infrastructure. However, because Subsystem 04 interfaces with *Plasma Ignition*, safety-critical operational modes must satisfy high-risk standards voluntarily.
Are we a GPAI model provider (Art. 53)?	**No.** We evaluate and audit open-weights alignment techniques. We do not place a foundation model on the EU market. However, GPAI systemic risk obligations (Art. 55) guide our capability-retention audits.
Is research and development exempt?	Yes, Art. 2(6) and Art. 2(8) exclude AI systems and models specifically developed and put into service for the sole purpose of scientific research and development prior to placing on the market.

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2. Why Mission 04 Directly Touches the AI Act

Preference optimization and activation steering directly address three mandatory pillars of the AI Act:

Article	Legal Obligation	Direct Connection to Mission 04
Art. 15(1)	High-risk AI systems must achieve appropriate levels of **accuracy, robustness, and cybersecurity**.	Unaligned base models exhibit sycophancy and hallucinations under stress. Measured on AURA Lab: DPO raises honest correction 48.5 % → 78.8 %, steering 48.5 % → 84.8 %, both without loss on arithmetic.
Art. 15(4)–(5)	Resilience against **errors and inconsistencies** and against **attempts to exploit vulnerabilities**.	Verbosity exploitation (length bias) is a form of reward hacking where models game evaluation metrics by outputting bloatware. We measured it (`03`): DPO doubled answer length; SimPO's length normalisation alone did not remove the bias on our data.
Art. 50(1)	**Transparency obligations** for systems interacting directly with natural persons.	Conversational systems must clearly disclose their AI nature and refrain from deceptive sycophantic behavior designed to manipulate human operators.

Article	Legal Obligation	Direct Connection to Mission 04
Art. 55(1)	GPAI models with systemic risk must undergo **adversarial testing and capability audits**.	The *Alignment Tax Audit* (`04`) measures where steering starts to cost capability (true-fact confirmation from α = 14, arithmetic from α = 17).

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3. Compliance Checklist for Mission 04

#	Requirement	Status	Verification & Evidence
1	**No personal data**	✓	AURA Lab uses an invented language (objects, colours, sums); no personal data is processed.
2	**Reproducibility**	✓	Fixed seeds in every script; the base model is cached; `05_Run_Log.txt` holds the full output; 19 tests pass.
3	**Results measured, not assumed**	✓	All reported numbers come from runs on the AURA Lab model (README, run log, `results/*.json`). Limits are stated: small model, one seed.
4	**Robustness against shortcuts (Art. 15)**	⚠ found & fixed	DPO trained only on "correct the user" pairs learned *always say no*: true facts confirmed 0 %. Adding "agree" pairs fixed it (100 %). Training data for AURA must contain both.
5	**Reward hacking via length (Art. 15)**	⚠ open	DPO doubled answer length (7 → 14 words) even without verbose training data. On length-confounded data SimPO did not remove the length bias (r = +0.67 vs +0.58 for DPO). Length must be monitored after any preference training.
6	**Steering safeguards**	✓	Measured window α ∈ [7, 9] at block 2. Over-steering (α ≥ 14) makes the model deny true facts; negative α makes it more sycophantic (12.1 % honest at α = −4). α must be clamped in code.
7	**Capability retention (Art. 55(1)(a))**	✓	Inside the window arithmetic stays at 100 % and perplexity rises only 1.479 → 1.502. The audit tracks three metrics so over-correction is visible.
8	**Calibration**	✓	Reward-model ECE 0.131 against a simulated rater panel; temperature scaling lowers it to 0.121 on held-out judgements.
9	**Energy / resources**	✓	SimPO removes the reference model: −50 % weight memory, −20 % measured training state, −14 % analytic for a 7B model.

4. Operational Protocols Before Deployment

1. **Clamp the steering coefficient** to the window measured for the deployed model and layer (here α ∈ [7, 9] at block 2); reject negative α. Re-measure after every model update.
2. **Protect the steering vector like model weights:** with the sign flipped it makes the model sycophantic (Art. 15(5), integrity against manipulation).
3. **Audit with three metrics** after every alignment change: target behaviour, its mirror (true-fact confirmation) and an unrelated capability.
4. **Check answer length** after preference training; length gains are a known form of reward hacking.
5. **Transparency (Art. 50):** AURA keeps telling users that they talk to an AI.
6. **Keep the evidence:** `05_Run_Log.txt`, `results/*.json` and the figures form the technical file for this subsystem.

Academic compliance assessment for BTU Cottbus-Senftenberg course submission. Not formal legal counsel.